D Y PATIL INTERNATIONAL UNIVERSITY
Sector 29, Pradhikaran, Akurdi, Pune - Maharashtra, INDIA 411044
(Establishment by Maharashtra Act No. LXIII of 2017)
School of Computer Science, Engineering & Applications

Course Name	Applied Computational Mathematics
Course Code	CSE1001
Credits	3
Prerequisites	Intermediate Mathematics
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. To introduce matrix methods and eigenvalue problems in linear algebra.
2. To develop techniques for analyzing and optimizing multivariable functions.
3. To present methods for solving ordinary differential equations.
4. To build understanding of integral and vector calculus in physical applications.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Linear Algebra – I - Matrices and matrix operations, Elementary row operations, Row echelon form, Reduced row echelon form, Gauss-Jordan elimination.	7
Module II: Linear Algebra – II - Rank and inverse of a matrix using row operations, Systems of homogeneous and non-homogeneous linear equations, Matrix equation $Ax = b$, Eigenvalues and eigenvectors.	7
Module III: Differential Calculus - Functions of several variables, Partial derivatives, Chain rule, Critical points, Maxima, Minima, Saddle points, Second derivative test, Lagrange multipliers.	7
Module IV: Differential Equations - Introduction to ordinary differential equations (ODEs), Order and degree of ODEs, First-order separable and linear DEs, Second-order linear homogeneous and nonhomogeneous DEs with constant coefficients, Method of variation of parameters.	7
Module V: Integral and Vector Calculus - Basic integration, Double integrals, Triple integrals, Applications in calculating area and volume, Gradient, Directional derivatives, Divergence, Curl, Physical interpretations.	8
Self Study/Concurrent Assessment Hours	9
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Apply matrix operations and row reduction techniques to solve systems of linear equations.
CO2	Analyze eigenvalues, eigenvectors, and matrix equations for solving engineering problems.
CO3	Solve multivariable calculus problems including optimization using partial derivatives and Lagrange multipliers.
CO4	Apply analytical methods to solve ordinary differential equations.
CO5	Apply integral and vector calculus concepts to compute area, volume, and analyze vector fields in physical systems.

CO-PO Mapping:			3: Substantial			2: Moderate					1: Slight		
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	1	-	-	-	-	-	-	-	-	1	2	1
CO2	3	3	2	-	-	-	-	-	-	-	-	1	3	2
CO3	3	3	2	-	-	-	-	-	-	-	-	1	3	2
CO4	3	2	1	-	-	-	-	-	-	-	-	1	2	2
CO5	3	3	2	1	-	-	-	-	-	-	-	1	3	2

TEXTBOOKS

1. Kreyszig, E. (2011). *Advanced Engineering Mathematics*. 10th ed. Wiley-India.
2. Grewal, B. S. (2014). *Higher Engineering Mathematics*. 43rd ed. Khanna Publishers.
3. Strang, G. (2016). *Introduction to Linear Algebra*. 5th ed. Wellesley-Cambridge Press.
4. Thomas, G. B., Weir, M. D., & Hass, J. (2014). *Thomas' Calculus: Early Transcendentals*. 13th ed. Pearson.

EVALUATION SCHEME

Continuous Assessment	Mid-term	End-term	Total
25	25	50	100

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School of Computer Science, Engineering & Applications

Course Name	Basic Electrical and Electronics Engineering
Course Code	CSE1002
Credits	3
Prerequisites	Basic knowledge of Physics,Elementary Mathematics
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. To understand the fundamental concepts of electrical circuits, sources, and network laws.
2. To explain and analyze the characteristics of semiconductor devices (diodes, BJTs, MOSFETs)
3. To apply and design rectifier and regulator circuits suitable forlow-power applications.
4. To construct, implement,and evaluate operational amplifier- based circuits for mathematical operations,signal conditioning, and processing.
5. To integrate and relate the concepts of electrical and electronic devices with computer engineering applications.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Fundamentals of Electrical Circuits Introduction to Electrical Engineering concepts, Circuit elements: Resistor, Inductor, Capacitor, Voltage and current sources (independent and dependent), Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL), Network theorems: Thevenin's, Norton's, and Superposition theorem, Source transformation, Signal waveforms: periodic and non-periodic	8
Module II: Semiconductor Devices and Applications Introduction to semiconductors (conductors, insulators, semiconductors), P–N junction diode: working principle and characteristics, Diode approximations, Rectifiers: Half-wave, Full-wave(center tap and bridge), Capacitor filter (qualitative analysis), Zener diode as voltage regulator (no-load and loaded regulator)	8
Module III: Bipolar Junction Transistors (BJT) and MOSFET BJT operation and characteristics, BJT as a switch and amplifier, Configurations: Common Base, Common Emitter, Common Collector, Biasing techniques and stability (conceptual level), Introduction to MOSFET:operation and types (n-channel,p-channel), Applications in switching and amplification	8

Module IV: Operational Amplifiers (Op-Amps) Introduction to differential amplifier, Ideal OP-AMP characteristics, Inverting and Non-inverting configurations, Applications of OP-AMP: Voltage follower,Adder and Subtractor, Integrator and Differentiator, Simple numerical problems	6
Module V: Applications in Computer Engineering Power supply design: Rectifier + Filter + Regulator integration, Role of diodes, transistors, and op-amps in digital systems, Amplifier circuits in computer hardware, Signal conditioning using Op-Amps in data acquisition systems,Overview of electronic circuits in computing devices (mother board, SMPS, memory circuits),Case studies and problem solving	6
Self Study/Concurrent Assessment Hours	9
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES At the end of the course the student will be able to:	
CO1	Apply basic circuit laws and network theorems to analyze simple electrical circuits.
CO2	Explain the working and characteristics of semiconductor devices
CO3	Design and analyze rectifier and regulator circuits for low-level power supply applications.
CO4	Implement and evaluate operational amplifier circuits
CO5	Relate electrical and electronic concepts to computer engineering applications

CO-PO Mapping:	3: Substantial			2: Moderate				1: Slight					
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	3	2	2	2	-	-	-	-	-	-	1	2	1
CO2	3	2	2	-	2	-	-	-	-	-	-	1	2	2
CO3	3	3	3	2	2	-	-	-	-	-	1	1	2	3
CO4	3	3	3	2	3	-	-	-	1	1	1	1	2	3
CO5	2	2	3	-	3	1	-	-	2	2	2	3	3	3

TEXTBOOKS

1. Jacob Millman, Christos C.Halkias, Integrated Electronics, McGrawHill.
2. Robert Boylestad and Louis Nashelsky, “Electronic Devices and Circuit Theory”, McGraw Hill Education (India) Private Limited, 2014.
3. William D.Cooper, “Electronic Instrumentation and Measurement Techniques”, PHI.

REFERENCES

1. David A.Bell, “Electronic Devices and Circuits”, Oxford University Press, 5th Edition, 2008.
2. D.P.Kothari, I. J.Nagrath, “Basic Electronics”, McGraw Hill Education (India) Private Limited, 2014.
3. A. K. Sawhney, “Electrical and Electronics measurements and instrumentation”, Dhanpat Rai and Co.

EVALUATION SCHEME

Continuous Assessment	Mid-term	End-term	Total
25	25	50	100

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Course Name	Computer Organization and Architecture
Course Code	CSE1003
Credits	3
Prerequisites	Fundamental concepts of computers, Basic programming concepts, Binary number system and logical thinking, Elementary understanding of digital systems
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. Understand the fundamental principles, evolution, and design concepts of computer organization and architecture.
2. Explain processor architecture, instruction execution, and performance improvement techniques.
3. Analyze memory hierarchy, cache organization, and memory performance.
4. Understand I/O mechanisms and parallel processing concepts in modern computer systems.
5. Apply architectural concepts to analyze system performance and design trade-offs.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Introduction to Computer Architecture & Data Representation Introduction to Computer Architecture: Evolution and history of computation, Computer architecture vs computer organization, Eight great ideas in computer architecture. Basic Computer Organization: Functional units of a computer, Block diagram of a computer system, CPU, memory. Programs, OS, and Hardware Interaction: Role of programs and compilers, Operating system interaction with hardware, Basic instruction execution concept. Number system.	9
Module II: Processor Architecture & Instruction-Level Parallelism Instruction Set Architecture (ISA): Concept of ISA, Types of instructions, Instruction formats (overview). Register Transfer Language and Microoperations: RTL notation, Register transfer and arithmetic microoperations. Processor Organization: ALU and its functions, Stack-based and general-purpose register architectures, Procedure calls and stack usage. RISC and CISC Architectures: Characteristics of RISC, Characteristics of CISC, Comparison.	8

Module III: Memory Systems & Cache Organization Memory Hierarchy: Concept and need for memory hierarchy, Levels of memory. Memory Performance: Latency, Throughput, Bandwidth. Memory Technologies: Overview of main memory technologies. Cache Memory: Purpose of cache memory, Instruction and data caches, Unified cache. Cache Mapping and Operations: Cache mapping techniques, Cache hit and miss, Write-through and write-back policies. Cache Performance: Tag and data arrays, Types of cache misses, Multi-level caches, Cache replacement policies (LRU).	8
Module IV: I/O Systems I/O Organization: I/O devices and controllers, I/O buses. I/O Techniques: Programmed I/O, Interrupts, Memory-mapped I/O, Direct Memory Access (DMA). Disk Systems: Disk structure and operation.	5
Module V: Multiprocessors and Parallel Processing Parallel Processing Fundamentals: Need for parallel processing, Speedup and performance metrics. Multiprocessor Systems: Shared-memory systems, Message-passing systems, Comparison of architectures. Parallel Programming and GPUs: Basics of parallel programming, GPU architecture concepts, Applications of GPUs.	6
Self Study/Concurrent Assessment Hours	9
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Explain the evolution of computer architecture, basic computer organization, interaction between programs–OS–hardware, and data representation using fixed-point and floating-point arithmetic..
CO2	Understand processor architecture including instruction sets, ALU and registers, RISC vs CISC, pipelining, hazards, and instruction-level parallelism techniques.
CO3	Explain memory hierarchy, memory technologies, cache organization, mapping and replacement policies, and analyze cache performance using hit and miss concepts.
CO4	Explain I/O system organization, data transfer mechanisms, interrupts, DMA, memory-mapped I/O, I/O buses, and disk systems used in modern computers.
CO5	Understand multiprocessor systems, performance metrics such as speedup, shared-memory and message-passing architectures, basic parallel programming concepts, and GPU architecture and applications.

CO-PO Mapping:	3: Substantial			2: Moderate						1: Slight				
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	-	-	-	-	-	-	-	-	-	2	3	2
CO2	3	3	2	2	2	-	-	-	-	-	-	2	3	3
CO3	3	3	2	2	2	-	-	-	-	-	-	2	3	3
CO4	3	2	2	2	2	-	-	-	-	-	-	2	2	3
CO5	3	3	2	2	2	-	-	-	-	-	-	2	3	3

TEXTBOOKS

1. Computer Organization and Architecture – William Stallings
2. Computer System Architecture – M. Morris Mano

3. Essentials of Computer Organization and Architecture – Null & Lobur

REFERENCES

1. Computer Architecture: A Quantitative Approach – Patterson & Hennessy
2. Introduction to Computing Systems: From Bits & Gates to C & Beyond – Patt & Patel
3. Code: The Hidden Language of Computer Hardware and Software – Charles Petzold

EVALUATION SCHEME

Continuous Assessment	Mid-term	End-term	Total
25	25	50	100

Course Name	Fundamentals of Web Technologies
Course Code	CSE1004
Credits	3
Prerequisites	Basic Computer Skills – familiarity with operating systems, file management, and use of internet browsers. Fundamentals of Programming – knowledge of variables, operators, loops and conditionals in languages such as C or Python. Problem-Solving Ability – logical reasoning and algorithmic thinking.
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. To introduce the fundamentals of the Internet and web technologies.
2. To provide skills in designing and developing responsive and accessible web pages.
3. To familiarize students with client-side and server-side scripting basics.
4. To expose students to modern web development frameworks and tools.
5. To align with industry trends in frontend, backend, and full-stack web development.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Introduction to Web and Internet Fundamentals Evolution of the Web - ARPANET, Web 1.0, ⋯, Web 4.0, AI in Web Architecture – client–server model, browser-server communication, DNS, domain names. Web protocols – HTTP/HTTPS, request-response cycle, cookies, sessions. Industry trends – cloud-native apps, semantic web, Web 3.0 and blockchain basics.	7
Module II: HTML5 and Web Page Structuring HTML basics – elements, tags, attributes, document structure. Text formatting and hyperlinks. Lists, tables, anchors, forms and input elements. HTML5 semantic tags – header, footer, nav, article, section. Embedding media – audio, video, iframes. Canvas and SVG basics. Responsive design with HTML meta viewport. Accessibility standards – ARIA roles, alt text, best practices.	7
Module III: Styling with CSS3 CSS syntax, selectors, colors, fonts. Box model and positioning – margin, padding, border, float layouts. Flexbox and CSS Grid. Advanced CSS – transitions, animations, pseudo-classes. Responsive design with media queries, mobile-first CSS. CSS frameworks – Bootstrap/Tailwind basics.	7

Module IV: Client-Side Scripting with JavaScript JavaScript basics – variables, data types, operators, control structures, loops, functions. Events and event handling – onClick, onChange, addEventListener. DOM manipulation – accessing and modifying elements. Form validation and real-time validation with JavaScript. ES6 features – let, const, arrow functions, template literals. Objects and arrays, JSON handling. AJAX and Fetch API, retrieving data from APIs. Mini project – To-do list or weather app using API.	7
Module V: Server-Side Fundamentals, Industry Tools, Trends and Deployment Introduction to server-side scripting. Node.js basics – setup, Hello World, npm packages. Express.js basics – routing, handling GET/POST requests. Databases in web – SQL vs NoSQL, MySQL/MongoDB overview. Connecting server to database, simple CRUD example. Industry trends – APIs, RESTful design, microservices. Version control with Git – git init, clone, commit, push. GitHub basics. Dev tools and debugging – browser console, inspect elements. Cloud deployment – Netlify, Vercel, Heroku. Web security basics – HTTPS, authentication, CORS, OWASP Top 10. Progressive Web Apps (PWA) – features, service workers, offline apps. Modern web trends – JAMstack, React/Angular overview, future directions.	8
Self Study/Concurrent Assessment Hours	9
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES At the end of the course the student will be able to:	
CO1	Explain concepts of the web, internet protocols, and architecture.
CO2	Design responsive web pages using HTML5, CSS3, and JavaScript.
CO3	Implement interactivity and data validation using client-side scripting.
CO4	Develop basic server-side applications, integrate databases, and understand REST APIs and cloud deployment concepts.
CO5	Use industry tools and modern technologies such as Git, browser developer tools, PWA concepts, and emerging web trends.

CO-PO Mapping:	3: Substantial			2: Moderate						1: Slight				
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	-	-	1	-	-	-	-	-	-	1	2	2
CO2	3	3	3	-	2	-	-	-	-	-	-	1	3	3
CO3	3	3	2	-	2	-	-	-	-	-	-	1	3	3
CO4	3	3	2	1	2	-	-	-	-	-	-	1	3	3
CO5	2	2	1	-	3	-	-	-	1	1	-	2	3	3

TEXTBOOKS

1. Jon Duckett – *HTML and CSS: Design and Build Websites* – Wiley.
2. Jon Duckett – *JavaScript and jQuery: Interactive Front-End Web Development* – Wiley.
3. Robin Nixon – *Learning PHP, MySQL and JavaScript* – O'Reilly.
4. Marty Stepp, Jessica Miller, Victoria Kirst – *Web Programming Step by Step* – Addison Wesley.

REFERENCES

1. David Flanagan – *JavaScript: The Definitive Guide* – O'Reilly.
2. Kyle Simpson – *You Don't Know JS* – O'Reilly.
3. Ethan Marcotte – *Responsive Web Design* – A Book Apart.
4. Bret Fisher, Michael Herman – *Full Stack Development with Node.js and Express.*
5. Chris Coyier – *CSS Secrets* – O'Reilly.

ONLINE REFERENCES

1. MDN Web Docs (developer.mozilla.org) – Authoritative documentation for HTML, CSS, and JavaScript.
2. W3C Standards (w3.org) – Official web standards and accessibility guidelines.
3. FreeCodeCamp (freecodecamp.org) – Tutorials and coding challenges.
4. W3Schools (w3schools.com) – Quick references for beginners.

EVALUATION SCHEME

Continuous Assessment	Mid-term	End-term	Total
25	25	50	100

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Course Name	Python Programming
Course Code	CSE1005
Credits	3
Prerequisites	Basic knowledge of computer fundamentals and C Programming
L-T-P-C	3-0-0-3

COURSE OBJECTIVES

1. Introduce Python as a powerful, high-level programming language for solving real-world computational problems.
2. Develop problem-solving skills using Python constructs, data structures, and libraries.
3. Enable students to apply Python in domains such as data processing, automation, and basic analytics.
4. Familiarize students with object-oriented programming concepts using Python.
5. Provide hands-on experience through laboratory work, mini projects, and a capstone project.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Introduction to Python: Features and applications of Python,Python installation and execution,Python syntax and indentation,Variables, data types, type conversion,Input/output statements	7
Module II:Control Structures and Functions : Conditional statements (if, if-else, nested if),Looping statements (for, while),Break, continue, pass,Functions: definition, parameters, return values,Lambda functions	7
Module III: Data Structures in Python : Strings and string operations,Lists and list methods,Tuples and sets,Dictionaries and dictionary operations,List comprehension, Regular Expression (Regex)	7
Module IV: Object Oriented Programming and File Handling : Classes and objects,Constructors and destructors,Inheritance, polymorphism, encapsulation,File handling (text and CSV files),Exception handling	7
Module V:Python Libraries and Applications: NumPy basics,Pandas for data analysis,Matplotlib for data visualization,Introduction to automation using Python,Case studies in real-world applications	8
Self Study/Concurrent Assessment Hours	9
Total Contact Hours (36+9)	**45**

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Understand and apply core Python programming concepts for problem solving.
CO2	Use Python data structures effectively for efficient data manipulation.
CO3	Implement modular and object-oriented programs in Python.
CO4	Apply Python libraries for file handling, data analysis, and visualization.
CO5	Design and develop real-world applications using Python through mini and capstone projects.

CO-PO Mapping:	3: Substantial			2: Moderate						1: Slight			
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	1	-	2	-	-	-	-	-	-	2	2	1
CO2	3	3	2	-	2	-	-	-	-	-	-	2	3	2
CO3	2	2	3	-	2	-	-	-	1	-	-	2	3	2
CO4	2	2	2	2	3	-	-	-	-	1	-	2	3	3
CO5	2	3	3	2	3	1	-	1	2	2	2	3	3	3

TEXTBOOKS

1. Reema Thareja, *John E. Freund's Mathematical Statistics with Applications* (8th edition).
2. Mark Lutz,*Learning Python, O'Reilly Media, Latest Edition.*

REFERENCES

1. Allen B. Downey, *Think Python: How to Think Like a Computer Scientist, O'Reilly Media.*
2. Charles R. Severance, *Python for Everybody: Exploring Data Using Python 3, Pearson Education.*

EVALUATION SCHEME

Continuous Assessment	Mid-term	End-term	Total
25	25	50	100

Course Name	IKS- Indian Science, Engineering and Technology
Course Code	CSE1006
Credits	1
Prerequisites	Some basic knowledge of Sanskrit, familiarity with Devnagri script, Indian number system will be useful
L-T-P-C	1-0-0-1

COURSE OBJECTIVES

1. To introduce the various streams of IKS.
2. To describe the contributions of ancient Indian researchers, scientists, and architects to the modern world.
3. To present the multi-dimensional nature of IKS and its significance in society.
4. To motivate the students to study IKS and explore its potential for overall development.

COURSE SYLLABUS

Modules	**Contact Hours**
Module I: Introduction to IKS - Introduction to the framework of the ancient Indian Traditional Knowledge and wisdom about natural heritage and climate. Introduction of the role of Vedas as the primary base of IKS. Relation of IKS to day-to-day life and modern-day knowledge systems.	3
Module II: Language and Art Studies in India - Understanding features of varnamala of Indian languages and learning about its origin, evolution, phonetic features and science of Paninian grammar briefly. Introduction to Ancient Indian music, dance and drama. Knowing the science of music and drama. Acknowledge the beauty of art, culture, folk music and traditional dance.	3
Module III: Indian Science and Technology - Discussing Mathematical references in Vedas. A brief development of Algebra, arithmetic, and geometry. Contribution of Brahmgupta, Acharya Pingala. Understanding astronomy as the science of calculating time, place, and direction. Learning important Ayurvedic lifestyle practices, water management, mining techniques, and modes of transportation in ancient India.	3
Module IV: Indian Agriculture and Economy -Learning the importance of agriculture in Human civilization. Historical review of sustainable farming in India. Understanding the Indian idea of well-organized polity, economy and administrative system.	3

Self Study/Concurrent Assessment Hours	3
Total Contact Hours (12+3)	**15**

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Develop an overall understanding of the various components of the Indian knowledge system.
CO2	Know the contribution of Indian Knowledge Systems to the world.
CO3	Apply traditional knowledge and techniques in day-to-day life.
CO4	Understand advanced study and interdisciplinary research on all aspects of the Indian knowledge system.

CO-PO Mapping:	3: Substantial			2: Moderate					1: Slight					
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	1	-	-	-	-	-	-	-	-	-	-	-
CO3	3	3	2	-	-	-	-	-	-	-	-	2	-	-
CO4	3	2	2	-	-	-	-	-	-	-	-	1	-	-

TEXTBOOKS

1. Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavan RN. (2022). *Introduction to Indian Knowledge System: Concepts and Applications*. PHI Learning Private Ltd.
2. Kapil Kapoor and A K Singh (2005). *Indian Knowledge Systems: Vol I and II*. D.K. Print World Ltd.
3. D. M. Bose, S. N. Sen and B. V. Subbarayappa (2010). *A Concise History of Science in India*. 2nd Ed., Universities Press, Hyderabad.
4. Dharampal (2021). *Indian Science and Technology in the Eighteenth Century: Some Contemporary European Accounts*. Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru.

EVALUATION SCHEME

Continuous Assessment	**Mid-term**	**End-term**	**Total**
25	25	50	100

Course Name	Basic Electrical and Electronics Engineering
Course Code	CSE1002P
Credits	1
Prerequisites	Basic knowledge of Physics,Elementary Mathematics
L-T-P-C	0-0-2-1

COURSE OBJECTIVES

1. To understand the fundamental concepts of electrical circuits, sources, and network laws.
2. To explain and analyze the characteristics of semiconductor devices (diodes, BJTs, MOSFETs)
3. To apply and design rectifier and regulator circuits suitable forlow-power applications.
4. To construct, implement,and evaluate operational amplifier- based circuits for mathematical operations,signal conditioning, and processing.
5. To integrate and relate the concepts of electrical and electronic devices with computer engineering applications.

LABORATORY

Sr. No.	Content
1	Verification of Ohm's Law–Determine the relationship between voltage, current, and resistance using a simple resistive circuit.
2	Verification of Kirchhoff's Laws (KVL and KCL)–Apply KVL in a series circuit and KCL in a parallel circuit, and verify through measurements.
3	Verification of Thevenin's Theorem– Reduce a given network into its Thevenin's equivalent circuit and compare measured vs calculated values
4	Verification of Norton's Theorem–Reduce a given network into its Norton's equivalent and validate using experimental setup.
5	Study of Half- Wave Rectifier– Observe input and output waveforms, and calculate ripple factor.
6	Study of Full- Wave Rectifier (Center-tap and Bridge)–Compare performance of full wave rectifier with half-wave rectifier.
7	Design and Testing of Zener Diode Voltage Regulator–Plot characteristics and verify regulation under load and no-load conditions.
8	Input-Output Characteristics of BJT (Common Emitter configuration)– Study transistor characteristics and calculate current gain.

9	BJT as a Switch–Demonstrate switching action of a transistor using LED or small load.
10	Applications of Operational Amplifier–Implement inverting, non- inverting, and voltage follower circuits, and verify performance.

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Apply basic circuit laws and network theorems to analyze simple electrical circuits.
CO2	Explain the working and characteristics of semiconductor devices
CO3	Design and analyze rectifier and regulator circuits for low-level power supply applications.
CO4	Implement and evaluate operational amplifier circuits
CO5	Relate electrical and electronic concepts to computer engineering applications

CO-PO Mapping:	3: Substantial			2: Moderate						1: Slight				
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	3	2	2	2	-	-	-	-	-	-	1	2	1
CO2	3	2	2	-	2	-	-	-	-	-	-	1	2	2
CO3	3	3	3	2	2	-	-	-	-	-	1	1	2	3
CO4	3	3	3	2	3	-	-	-	1	1	1	1	2	3
CO5	2	2	3	-	3	1	-	-	2	2	2	3	3	3

TEXTBOOKS

1. Jacob Millman, Christos C.Halkias, Integrated Electronics, McGrawHill.
2. Robert Boylestad and Louis Nashelsky, “Electronic Devices and Circuit Theory”, McGraw Hill Education (India) Private Limited, 2014.
3. William D.Cooper, “Electronic Instrumentation and Measurement Techniques”, PHI.

REFERENCES

1. David A.Bell, “Electronic Devices and Circuits”, Oxford University Press, 5th Edition, 2008.
2. D.P.Kothari, I. J.Nagrath, “Basic Electronics”, McGraw Hill Education (India) Private Limited, 2014.
3. A. K. Sawhney, “Electrical and Electronics measurements and instrumentation”, Dhanpat Rai and Co.

EVALUATION SCHEME

Continuous Assessment			**End-term Practical**	**Total**
Lab Manual	**Lab Assessment**	**Internal Viva-voce**		
10	10	30	50	100

Course Name	Fundamentals of Web Technologies Practical
Course Code	CSE1004P
Credits	1
Prerequisites	Basic Computer Skills – familiarity with operating systems, file management, and use of internet browsers. Fundamentals of Programming – knowledge of variables, operators, loops and conditionals in languages such as C or Python. Problem-Solving Ability – logical reasoning and algorithmic thinking.
L-T-P-C	0-0-2-1

COURSE OBJECTIVES

1. To introduce the fundamentals of the Internet and web technologies.
2. To provide skills in designing and developing responsive and accessible web pages.
3. To familiarize students with client-side and server-side scripting basics.
4. To expose students to modern web development frameworks and tools.
5. To align with industry trends in frontend, backend, and full-stack web development.

LABORATORY

Sr. No.	Content
1.	Basic HTML Page Design
2.	Lists, Tables, and Forms in HTML
3.	HTML5 Semantic Web Page with Multimedia
4.	Web Page Styling using CSS3
5.	Layout Design using Flexbox and CSS Grid
6.	Responsive Web Design using Media Queries and Bootstrap/Tailwind
7.	JavaScript DOM Manipulation and Form Validation
8.	Mini Project – Dynamic Web Application (To-do List / Weather App)

LAB COURSE OUTCOMES

At the end of the course the student will be able to:

CO1	Design and develop basic web pages using HTML5 elements, lists, tables, and forms.

CO2	Apply CSS3 techniques including layout design, styling, Flexbox, and Grid to create visually appealing web pages.
CO3	Develop responsive web interfaces using media queries and modern frameworks such as Bootstrap or Tailwind.
CO4	Implement client-side interactivity using JavaScript including DOM manipulation and form validation.
CO5	Build and deploy dynamic web applications integrating HTML, CSS, JavaScript, and APIs.

CO-PO Mapping:			3: Substantial				2: Moderate					1: Slight		
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	2	2	1	2	-	-	-	-	2	-	2	2	2
CO2	3	2	3	2	3	-	-	-	-	2	-	2	3	3
CO3	3	2	3	2	3	-	-	-	-	2	-	2	3	3
CO4	3	3	3	2	3	-	-	-	-	2	-	2	3	3
CO5	3	3	3	3	3	-	-	-	-	2	-	2	3	3

TEXTBOOKS

1. Jon Duckett – *HTML and CSS: Design and Build Websites* – Wiley.
2. Jon Duckett – *JavaScript and jQuery: Interactive Front-End Web Development* – Wiley.
3. Robin Nixon – *Learning PHP, MySQL and JavaScript* – O'Reilly.
4. Marty Stepp, Jessica Miller, Victoria Kirst – *Web Programming Step by Step* – Addison Wesley.

REFERENCES

1. David Flanagan – *JavaScript: The Definitive Guide* – O'Reilly.
2. Kyle Simpson – *You Don't Know JS* – O'Reilly.
3. Ethan Marcotte – *Responsive Web Design* – A Book Apart.
4. Bret Fisher, Michael Herman – *Full Stack Development with Node.js and Express.*
5. Chris Coyier – *CSS Secrets* – O'Reilly.

ONLINE REFERENCES

1. MDN Web Docs (developer.mozilla.org) – Authoritative documentation for HTML, CSS, and JavaScript.
2. W3C Standards (w3.org) – Official web standards and accessibility guidelines.
3. W3Schools (w3schools.com) – Quick references for beginners.

EVALUATION SCHEME

Continuous Assessment			**End-term Practical**	**Total**
Lab Manual	**Lab Assessment**	**Internal Viva-voce**		
10	10	30	50	100

Course Name	Python Programming
Course Code	CSE1005P
Credits	1
Prerequisites	Basic knowledge of computer fundamentals
L-T-P-C	0-0-2-1

COURSE OBJECTIVES

1. Develop practical programming skills using Python constructs, control structures, and functions.
2. Apply Python data structures and regular expressions for solving computational problems.
3. Implement object-oriented programming, file handling, and exception handling concepts using Python.
4. Use Python libraries for data analysis and visualization.

LABORATORY

Sr. No.	Content
1	Implementation of Looping Constructs for Series Generation and Number Manipulation
2	Implementation of Conditional Statements for Decision-Making Problems
3	Design and Implementation of User-Defined Functions and Lambda Expressions
4	Implementation of Python Data Structures Using Lists, Tuples, Sets, and Dictionaries
5	Pattern Matching and Validation Using Regular Expressions in Python
6	Design and Implementation of Object-Oriented Programming Concepts in Python
7	Implementation of File Handling and Exception Handling Mechanisms in Python
8	Data Analysis and Visualization Using NumPy, Pandas, and Matplotlib Libraries

COURSE OUTCOMES	
At the end of the course the student will be able to:	
CO1	Implement Python programs using looping constructs, conditional statements, and functions.
CO2	Apply Python data structures and regular expressions to solve computational problems.
CO3	Design and implement object-oriented programs using Python.
CO4	Implement file handling and exception handling mechanisms in Python applications.
CO5	Perform data analysis and visualization using NumPy, Pandas, and Matplotlib libraries.

CO-PO Mapping:	3: Substantial			2: Moderate						1: Slight			
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
CO1	3	3	2	2	3	-	-	-	-	-	-	-	2	2
CO2	3	3	3	2	3	-	-	-	-	-	-	-	3	2
CO3	3	3	3	2	3	-	-	-	-	-	-	-	3	2
CO4	3	3	2	3	3	-	-	-	-	-	-	-	3	2
CO5	3	3	3	3	3	-	-	-	-	-	-	-	3	3

TEXTBOOKS

1. Reema Thareja, *John E. Freund's Mathematical Statistics with Applications* (8th edition).
2. Mark Lutz, *Learning Python, O'Reilly Media, Latest Edition.*

REFERENCES

1. Allen B. Downey, *Think Python: How to Think Like a Computer Scientist, O'Reilly Media.*
2. Charles R. Severance, *Python for Everybody: Exploring Data Using Python 3, Pearson Education.*

EVALUATION SCHEME

Continuous Assessment			**End-term Practical**	**Total**
Lab Manual	**Lab Assessment**	**Internal Viva-voce**		
10	10	30	50	100